

Atharva Hule

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EDUCATION

Sardar Patel Institute of Technology (CGPA: 7.52)

Bachelor of Technology in Computer Science and Engineering

Andheri, Mumbai

Sept. 2024 – Present

Pace Junior Science College (88.83%)

11th-12th Science

Nerul, Navi Mumbai

Aug. 2022 – May 2024

Father Agnells School (93.00%)

11th-12th Science

Vashi, Navi Mumbai

Sept. 2016 – Apr 2022

Birla Public School

11th-12th Science

Doha, Qatar

July 2009 – Aug 2016

PROJECTS

Solar Power Prediction System | *React, Node.js, Express, FastAPI, Python, MongoDB, Scikit-learn*

- Built a full-stack solar power prediction platform integrating React, Node.js/Express, FastAPI, MongoDB, and a Random Forest regression model for solar output forecasting
- Integrated weather and solar irradiance data including ambient temperature, module temperature, wind speed, and irradiance to generate location-based power predictions
- Developed an interactive dashboard supporting current and future date/time predictions, data visualization, prediction history, filtering, sorting, and CSV export
- Implemented JWT-based authentication and REST APIs connecting the frontend, backend, database, and ML inference service
- Integrated a Groq-powered AI assistant for natural-language interaction with prediction data and optimized the trained ML model for efficient inference

Green-Finance: Real-Time Carbon & Energy Observability for Financial Microservices | *FastAPI, Docker, Kepler (eBPF), Prometheus, Grafana*

- Proposing and designing a team-based observability framework to monitor real-time energy consumption and carbon footprint of financial microservices (payment processing, authentication)
- Architecting modular banking logic in FastAPI, containerized with Docker, integrated with Kepler for kernel-level (eBPF-based) energy telemetry
- Applying Data Structures and DBMS concepts for efficient telemetry data handling and historical carbon auditing
- Leveraging Computer Networks and Operating Systems concepts to manage microservice communication and kernel-level energy tracking
- Building custom Grafana dashboards on Prometheus-aggregated data to report a "Carbon-per-Transaction" sustainability metric

TECHNICAL SKILLS

Languages: Java, Python, MySQL, JavaScript, HTML/CSS

Frameworks: React, Node.js, FastAPI

Developer Tools: Git, VS Code, PyCharm, IntelliJ

INTERESTS

- Problem Solving
- Exploring Emerging Technologies
- Football